Basic Ciphers Explained with Examples

1. Caesar Cipher

Description: A substitution cipher that shifts each letter in the plaintext by a fixed number of places down the alphabet.

Example:

Plaintext: HELLO

Key (Shift): 3

Ciphertext: KHOOR

2. Vigenère Cipher

Description: A polyalphabetic substitution cipher that uses a keyword to determine the shift for each letter.

Example:

Plaintext: ATTACKATDAWN

Keyword: LEMON (Repeated: LEMONLEMONLE)

Ciphertext: LXFOPVEFRNHR

3. Playfair Cipher

Description: A digraph substitution cipher using a 5x5 grid of letters.

Example:

Keyword/Grid: MONARCHY

Plaintext: HELLO

Ciphertext: CFVT

4. Transposition Cipher

Description: Rearranges the characters in the plaintext according to a specific pattern.

Example:

Plaintext: ATTACKATDAWN

Key: 4 (Columns)

Ciphertext: ACTADTKAWATN

5. Hill Cipher

Description: A polygraphic cipher that uses matrix multiplication to encrypt plaintext.

Example:

Key Matrix: [2, 3; 1, 4]

Plaintext: HI

Ciphertext: RL

6. Substitution Cipher

Description: Each letter in the plaintext is replaced with a different letter from a substitution alphabet.

Example:

Substitution Alphabet: DCBAGHIJKLMNOPQRSTUVWXYZEF

Plaintext: HELLO

Ciphertext: EBJJI

7. Vernam Cipher (One-Time Pad)

Description: A perfect cipher where the plaintext is XORed with a random key of the same length.

Example:

Plaintext (Binary): 1011

Key (Binary): 1100

Ciphertext: 0111

8. Rail Fence Cipher

Description: A transposition cipher where text is written diagonally in rows and then read row by row.

Example:

Plaintext: ATTACKATDAWN

Key: 3 Rows

Ciphertext: AKDTAAATCN

9. Affine Cipher

Description: A substitution cipher where each letter is encoded using a linear function.

Example:

Plaintext: HELLO

Key: a = 5, b = 8

Ciphertext: RCLLA